

Knowns:

$$F = 600 \text{ lbf}$$

$$\delta_{\text{allow}} = 0.005$$

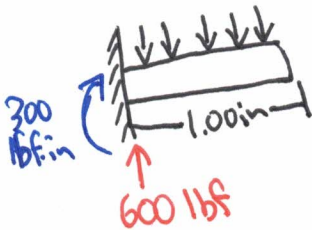
$$\sigma_{\text{allow}} = \frac{S_y}{SF} = \frac{40,000}{4} = 10,000$$

$$SF = 4$$

Material: 6061 T6 aluminum

$$E = 10,000,000 \text{ psi} \quad S_y = 40,000 \text{ psi}$$

Feature A:



Known:

$$W = 600 \text{ lbf}, \quad w = 600 \frac{\text{lbf}}{\text{in}}$$

$$L = 1.00 \text{ in}$$

$$M_{\text{max}} = \frac{WL}{2} = \frac{600(1)}{2} = 300 \text{ lbf}\cdot\text{in}$$

$$I = \frac{\pi d^4}{64}, \quad \sigma = \frac{M_c}{I} = \frac{32M}{\pi d^3}$$

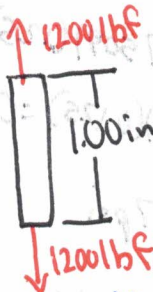
$$d_{\text{strength}} = \sqrt[3]{\frac{32M}{\pi \sigma_{\text{allow}}}} = \sqrt[3]{\frac{32(300)}{\pi (10,000)}} = 0.6$$

$$\delta = \frac{WL^2}{8EI} = \frac{8WL^3}{E\pi d^4}$$

$$\text{nominal } d = 0.750 \text{ in}$$

$$d_{\text{stiffness}} = \sqrt[4]{\frac{8WL^2}{E\pi \delta_{\text{allow}}}} = \sqrt[4]{\frac{8(600)(1)^2}{10^7 \pi (0.005)}} = 0.418 \text{ in}$$

Feature B:



Knowns

$$P = 1200 \text{ lbf}$$

$$b = 0.500 \text{ in}$$

$$L = 1.00 \text{ in}$$

unknown:

thickness: t

Stiffness:

$$\delta = \frac{PL}{Eb^3t}$$

$$t_{\text{stiffness}} = \frac{PL}{Eb\delta_{\text{allow}}}$$

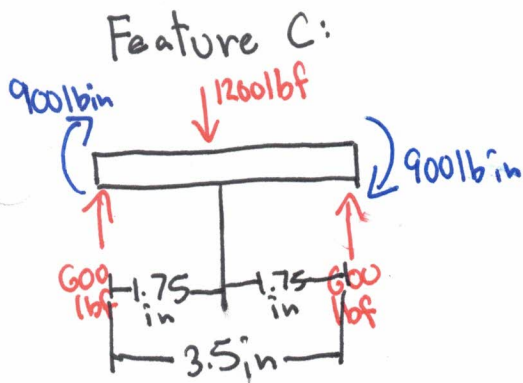
$$= \frac{1200(1)}{10^7(0.5)(0.005)} = 0.0480 \text{ in}$$

$$\text{select } t = 0.250 \text{ in}$$

Strength

$$A = bt, \quad \sigma = \frac{P}{bt}$$

$$t_{\text{strength}} = \frac{P}{b\sigma_{\text{allow}}} = \frac{1200}{0.5(10,000)} = 0.24 \text{ in}$$



Knowns:

$$P = 1200 \text{ lbf} \quad L = 3.50 \text{ in} \quad b = 1.00 \text{ in}$$

$$R_1 = R_2 = 600 \text{ lbf}$$

$$M_0 = 600(1.5) = 900 \text{ lbf}\cdot\text{in}$$

$$M(x) = -900 + 600x$$

$$I = \frac{bt^3}{12}, \quad \sigma = \frac{6M_{\max}}{bt^2}$$

$$M_{\max} = 900 \text{ lbf}\cdot\text{in}$$

$$t_{\text{strength}} = \sqrt{\frac{6M_{\max}}{b\sigma_{\text{allow}}}} = \sqrt{\frac{6(900)}{1(10000)}} = 0.7348 \text{ in}$$

$$EI_y = \frac{Px(3L^2 - 4x^2)}{48} - \frac{M_0x(L-x)}{2}$$

$$EI_y = -686.25x + 450x^2 - 100x^3$$

$$EI_y' = -686.25 + 900x - 300x^2 = 0$$

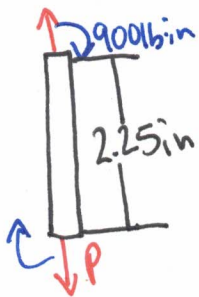
$$x = 1.25, 1.75 \text{ in}$$

$$\max EI_y = 312.5 \text{ lb}\cdot\text{in}^3$$

$$\delta_{\max} = \frac{312.5}{EI} = \frac{3750}{Ebt^3}$$

$$t_{\text{stiffness}} = \sqrt[3]{\frac{3750}{Eb\delta_{\text{allow}}}} = \sqrt[3]{\frac{3750}{10^7(1)(0.005)}} = 0.4217 \text{ in}$$

Feature D:



$$N = 600 \text{ lbf} \quad H = 2.25 \text{ in}$$

$$M = 900 \text{ lbf}\cdot\text{in} \quad h = 1.00 \text{ in}$$

$$A = ht, \quad I = \frac{th^3}{12}, \quad \sigma_{\max} = \frac{N}{ht} + \frac{6M}{th^2}$$

$$t_{\text{strength}} = \frac{N/h + \frac{6M}{h^2}}{\sigma_{\text{allow}}} = \frac{600 + \frac{6(900)}{1^2}}{10000} = 0.6000 \text{ in}$$

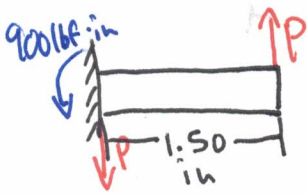
$$\delta_{\text{axial}} = \frac{NH}{Et} = \frac{600(2.25)}{E(1)} = 0.000135$$

$$\delta_{\text{lateral}} = \frac{MH^2}{2EI} = \frac{6MH^2}{Eth^3} = \frac{0.0027337}{1}$$

$$\delta = \sqrt{\delta_{\text{axial}}^2 + \delta_{\text{lateral}}^2}$$

$$t_{\text{stiffness}} = \frac{\sqrt{(0.000135)^2 + (0.0027337)^2}}{0.005} = 0.5474 \text{ in}$$

Feature E:



Known:

$$P = 600 \text{ lbf} \quad b = 1.00 \text{ in} \\ L = 1.5 \text{ in}$$

Unknown:

vertical thickness t

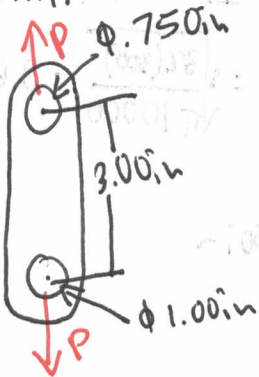
$$M_{\max} = PL = 600(1.5) = 900 \text{ lbf in}$$

$$\sigma = \frac{6PL}{bt^2}$$

$$t_{\text{strength}} = \sqrt{\frac{6PL}{b\sigma_{\text{allow}}}} = \sqrt{\frac{6(600)(1.5)}{1(10,000)}} = 0.7348 \text{ in}$$

$$t_{\text{stiffness}} = \sqrt[3]{\frac{4PL^3}{Eb\delta_{\text{allow}}}} = \sqrt[3]{\frac{4(600)(1.5)^3}{10^7(1)(0.005)}} = 0.5451 \text{ in}$$

Link:



$$P = 600 \text{ lbf} \quad w = 1.5 \text{ in}$$

$$L = 3.00 \text{ in}$$

$$L_{\text{overall}} = L + 2r = 3 + 2(0.75) = 4.5 \text{ in}$$

$$d_{\text{amax}} = 0.7508 \quad d_{\text{imin}} = 1.0005 \text{ in}$$

$$A_{\text{req}} = \frac{F}{\sigma_{\text{allow}}} = \frac{600}{10000} = 0.06 \text{ in}^2$$

$$A_{\text{net}} = (w - d)t$$

$$t_A = \frac{0.06}{1.5 - 0.7508} = 0.08009 \text{ in}$$

$$t_i = \frac{0.06}{1.5 - 1.0005} = 0.12012 \text{ in}$$

$$\sigma_i = \frac{600}{0.062} = 9610 \text{ psi}$$

$$\sigma_A = \frac{600}{0.09365} = 6407 \text{ psi}$$

$$n_{\min} = \frac{40000}{960961} = 4.16$$

Stiffness:

$$A_{\text{req}} = \frac{FL}{E\delta_{\text{allow}}} = \frac{600(3)}{10^7(0.005)} = 0.0360 \text{ in}^2$$

$$t_{\text{stiffness}} = \frac{0.036}{1.5 - 1.0005} = 0.07207 \text{ in}$$

$$\sum \frac{FL}{EA_{\min}} = \frac{600(3)}{10^7(0.0624)} = 0.002883$$

$$L_{\max} = \frac{\delta_{\text{allow}} EA_{\min}}{F} = \frac{0.005(10^7)(0.0624)}{600} = 5.203 \text{ in}$$